

PROBLEM STATEMENT

To design and develop a cross-modal deepfake detection system that integrates speech and lip movement analysis to accurately identify manipulated audio-visual content.

Real World Impact

- Political misinformation and election manipulation
- Identity fraud in banking and KYC verification
- Misuse of synthetic media and impersonation
- Loss of trust in digital news and evidence

PROJECT OBJECTIVES

The primary objectives of this project are:

- Develop a cross-modal deepfake detection system using both audio and video analysis.
- Detect manipulated videos using lip-region and visual feature analysis.
- Improve cross-dataset generalization using unseen benchmark datasets.

SYSTEM DESIGN/ARCHITECTURE

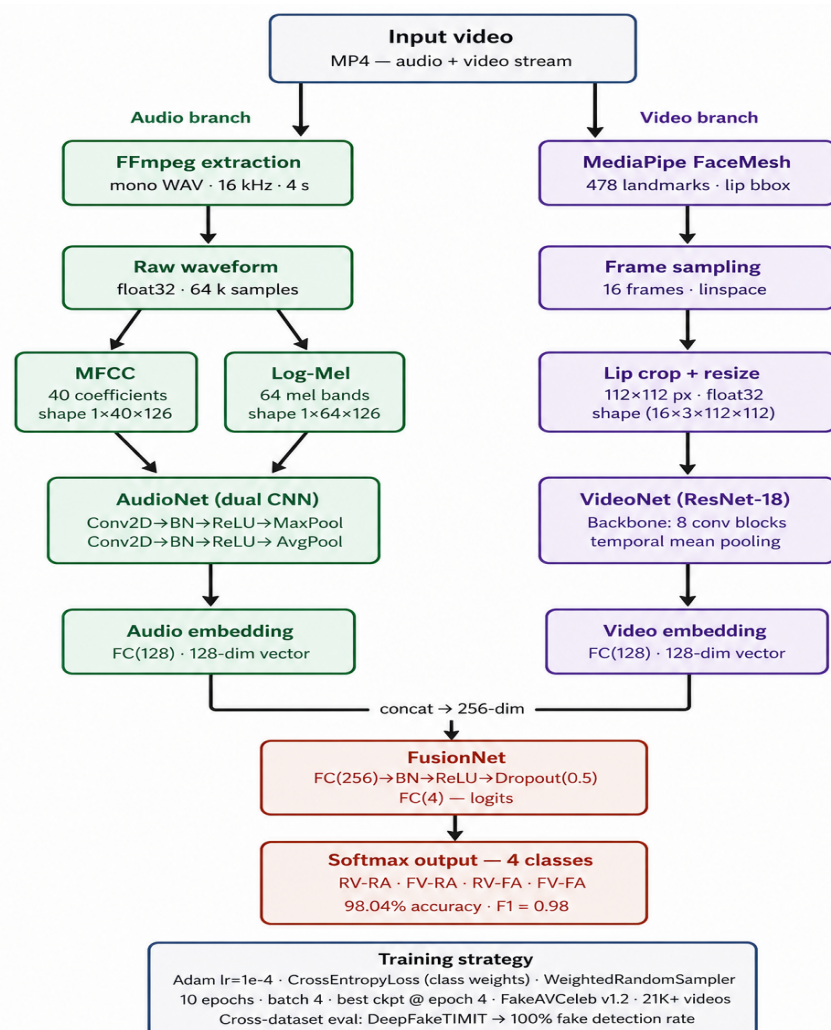


Fig. 1: The proposed multimodal Deepfake Detection Architecture

METHODOLOGY

- Parallel Multimodal Processing:** The proposed system processes audio and video streams simultaneously for robust deepfake detection.
- Audio Feature Extraction:** Audio is converted into MFCC and Log-Mel spectrogram features, which are encoded using a dual CNN-based AudioNet to generate a 128-dimensional embedding.
- Visual Feature Extraction:** Lip-region frames extracted using MediaPipe FaceMesh are processed through a ResNet-18 based VideoNet to produce a 128-dimensional visual embedding.
- Fusion & Classification:** Audio and video embeddings are concatenated into a 256-dimensional feature vector and passed through a fusion network for 4-class deepfake classification (RV-RA, RV-FA, FV-RA, FV-FA).

IMPLEMENTATION DETAILS

Framework	PyTorch 2.5.1 + CUDA 12
Audio	Librosa 0.11 FFmpeg
Video	OpenCV 4.11 MediaPipe 0.10.9
Backbone	ResNet-18 (ImageNet pretrained)
Hardware	NVIDIA RTX 4060 Laptop GPU 8 GB
OS / Env	Windows 11 + Anaconda Python 3.9
Optimizer	Adam lr=1e-4 10 epochs
Batch	Size 4 WeightedRandomSampler

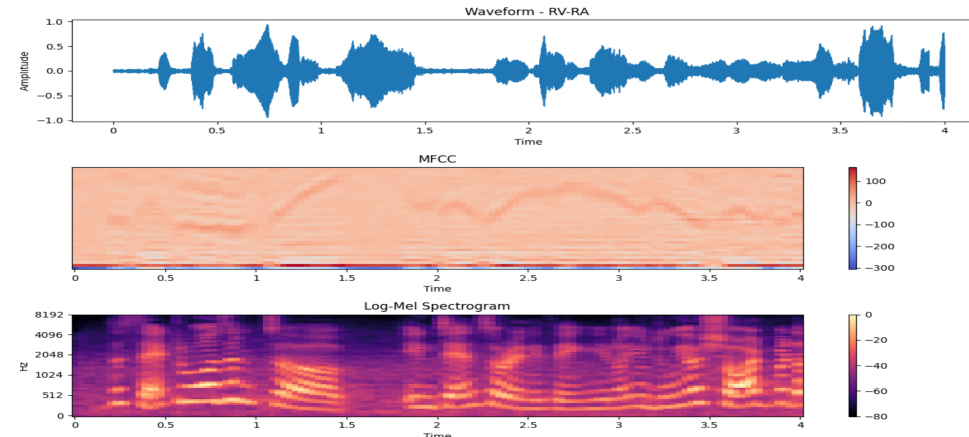


Fig. 2: Audio Feature extraction: waveform, MFCC and Log-Mel Spectrogram

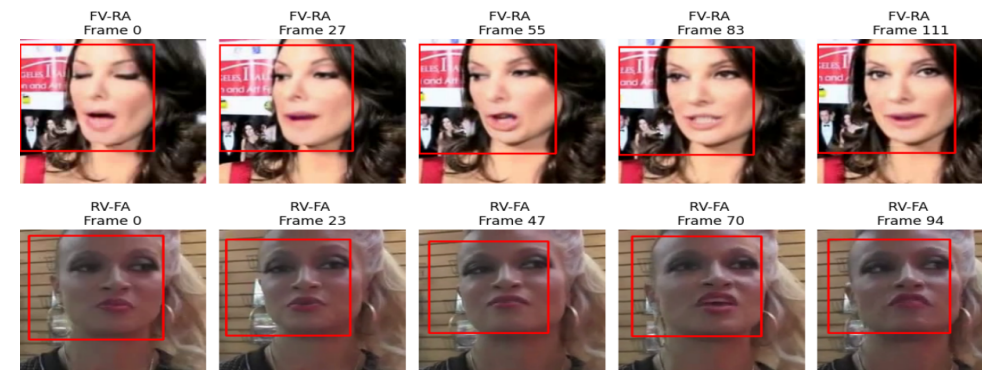


Fig. 3: Lip region detection applied on video frames to isolate mouth movements.

RESULTS AND PERFORMANCE

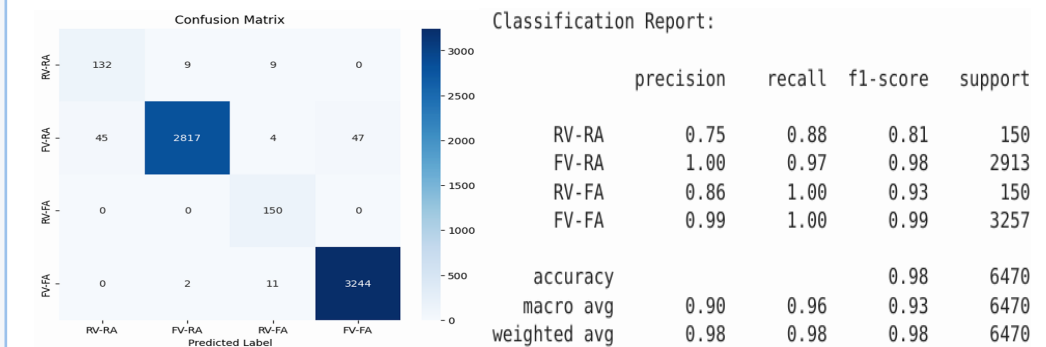


Fig. 4: Confusion Matrix of the AVFusion Net Model

Fig. 5: Classification Report of the Model Showing High Precision, Recall and F1 Scores.

The proposed AVFusionNet model achieved an accuracy of 98.04% on FakeAVCeleb dataset, showing very low misclassification across classes, demonstrating effective multimodal fusion of audio and video features.

CROSS-DATASET GENERALIZATION

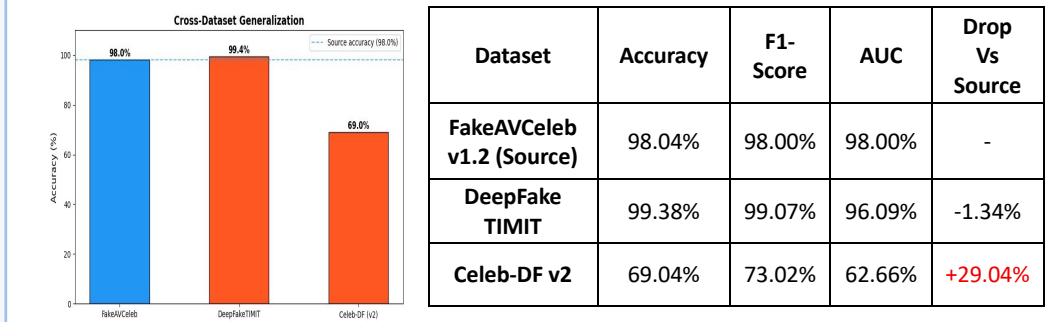


Fig. 6: Comparison of Model Accuracy across different datasets

- The trained AVFusionNet model was evaluated on unseen external datasets to assess generalization capability.
- Results indicate that the model achieves high accuracy on the source dataset (FakeAVCeleb) and DeepFakeTIMIT, while a noticeable drop is observed on Celeb-DF v2 due to domain differences and increased data complexity.

CONCLUSION & FUTURE SCOPE

AVFusionNet achieved an accuracy of 98.04% on FakeAVCeleb dataset using effective audio-video fusion and 4 class classification enabled and improved fine grained manipulation detection performance. Cross-dataset testing demonstrated strong generalization capability while highlighting challenges due to domain shift.

Future work includes improving generalization capability of the model with diverse datasets, adding Explainable AI, using attention based fusion and increasing robustness against advanced deepfakes.

REFERENCES (IEEE)

- [1] M. Javed *et al.*, "Audio-Visual Synchronization and Lip Movement Analysis for Real-Time Deepfake Detection," *Int. J. Comput. Intell. Syst.*, vol. 18, p. 170, 2025.
- [2] H. Khalid *et al.*, "FakeAVCeleb: A novel audio-video multimodal deepfake dataset," *arXiv preprint arXiv:2108.05080*, 2021.
- [3] D. Salvi *et al.*, "A Robust Approach to Multimodal Deepfake Detection," *J. Imaging*, vol. 9, p. 122, 2023.